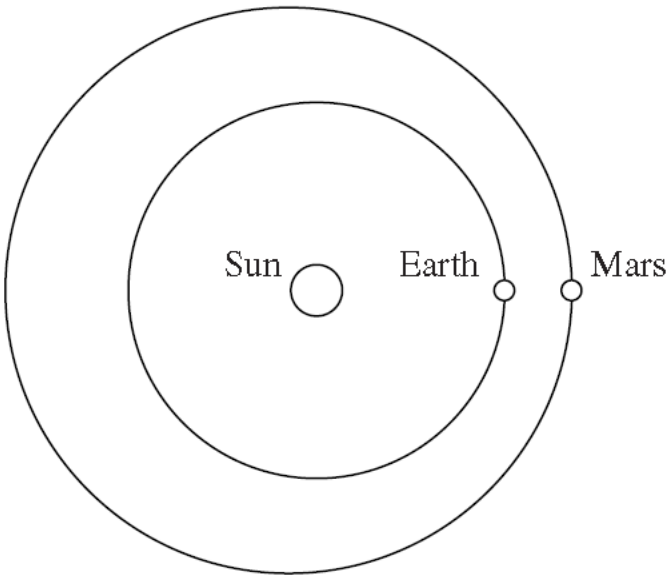


3	(a)	(i)	Explain what is meant by the term <i>escape speed</i> .	
				
				
			<u>minimum velocity that the body must have at the Earth's surface in</u>	
				
			<u>order to escape from the influence of the Earth's gravitational field</u>	
				[2]
		(ii)	Mars has a radius of approximately 3.4×10^6 m and a mass of 6.4×10^{23} kg. Show that the escape speed from Mars is approximately 5 km s^{-1} .	
			gain in gravitational potential energy in reaching infinity is $\Delta E_p = 0 - \left(-\frac{GMm}{r}\right) = \frac{GMm}{r}$ minimum kinetic energy required is equal to the gain in gravitational potential energy in reaching infinity: $\frac{1}{2} mv_{\min}^2 = \frac{GMm}{r}$ $v_{\min} = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2(6.67 \times 10^{-11})6.4 \times 10^{23}}{3.4 \times 10^6}} = 5011$ $\approx 5 \text{ km s}^{-1}$	
				[3]

		(iii)	Suggest why a rocket would be able to escape from Mars with an initial speed much less than the escape speed given in part (a)(ii).																					
			can be done at any speed given sufficient fuel																					
																								
			OR differentiates from escape speed – unpowered etc \																					
			OR fuel/chemical energy																					
				[1]																				
	(b)	Fig. 3.1 shows the Sun, Earth and Mars in alignment. Earth and Mars rotate around the Sun in the same directional sense.  Not to scale Fig. 3.1 A rocket of mass 2.05×10^6 kg leaves the surface of Mars closest to Earth and heads for Earth. Fig. 3.2 below gives data relevant to the rocket at the start of its journey. <table border="1" data-bbox="323 1462 1399 1821"> <thead> <tr> <th>astronomical object (AO)</th><th>mass of AO / kg</th><th>distance of rocket from the centre of AO / m</th><th>rocket's gravitational potential due to AO / J kg⁻¹</th><th>sign of gravitational potential</th></tr> </thead> <tbody> <tr> <td>Mars</td><td>6.4×10^{23}</td><td>3.4×10^6</td><td>1.26×10^7</td><td></td></tr> <tr> <td>Earth</td><td>6.0×10^{24}</td><td>5.6×10^{10}</td><td></td><td>negative</td></tr> <tr> <td>Sun</td><td>2.0×10^{30}</td><td>2.3×10^{11}</td><td>5.80×10^6</td><td></td></tr> </tbody> </table> Fig. 3.2			astronomical object (AO)	mass of AO / kg	distance of rocket from the centre of AO / m	rocket's gravitational potential due to AO / J kg ⁻¹	sign of gravitational potential	Mars	6.4×10^{23}	3.4×10^6	1.26×10^7		Earth	6.0×10^{24}	5.6×10^{10}		negative	Sun	2.0×10^{30}	2.3×10^{11}	5.80×10^6	
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		(i)	Complete Fig. 3.2 by calculating the gravitational potential of the rocket due to the presence of Earth and the signs of the gravitational potential energies due to Mars itself and the Sun.	
			$\phi = -\frac{GM}{r} = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{5.6 \times 10^{10}}$ $= -7.15 \times 10^3 \text{ J kg}^{-1}$ (formula provided hence no mark) negative, negative	
				[2]
		(ii)	Calculate the total gravitational potential energy of the rocket on the surface of Mars.	
			As rocket is at surface of Mars, i.e. start of journey, and there is potential due to all 3 AOs, thus total GPE of rocket = $m(\Sigma\phi)$ $= 2.05 \times 10^6 (-5.80 \times 10^6 + -1.26 \times 10^7 + -7.15 \times 10^3)$ $= -3.77 \times 10^{13} \text{ (ecf from table)}$	
			total gravitational potential energy = J	[2]
		(c) (i)	Derive an expression to show that for satellites in a circular orbit	
			$T^2 \propto r^3$ where T is the period of orbit and r is the radius of the orbit.	
			LO9c gravitational force provides centripetal force $\frac{GM_s m}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$ $\Rightarrow T^2 = \frac{4\pi^2}{GM_s} r^3$ $\Rightarrow T^2 \propto r^3$	
				[2]
		(ii)	The orbits of the Earth and Mars can be approximated to be circular orbits around the Sun.	
			Hence, estimate the orbital period of Mars.	
			$T^2 \propto r^3$ $\frac{T_{Mars}^2}{365^2} = \frac{(2.3 \times 10^{11} + 3.4 \times 10^6)^3}{(2.3 \times 10^{11} - 5.6 \times 10^{10})^3}$ $T = 554 \text{ days} \approx 600 \text{ days (1 s.f.)}$	
			orbital period of Mars = days	[3]

4	A fixed mass of ideal gas at a low temperature is trapped in a container at constant pressure
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